

Mouse Corneal Immune Cell Heterogeneity Revealed by Single-Cell RNA Sequencing

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PURPOSE. This study aimed to define the heterogeneity, spatial localization, and functional roles of immune cells in the mouse cornea using single-cell RNA sequencing (scRNA-seq) and immunofluorescent staining.

METHODS. Enriched mouse corneal immune cells (C57BL/6 strain, age 16–20 weeks) underwent single-cell RNA sequencing library preparation, sequencing, and analysis with Seurat, Monocle 3, and CellChat packages in R. Pathway analysis used Qiagen Ingenuity Pathway Analysis software. Immunostaining confirmed cell distribution.

RESULTS. We identified 14 distinct immune cell clusters (56% myeloid and 44% lymphoid). Myeloid populations included resident macrophages, conventional dendritic cells (cDC2s), Langerhans cells, neutrophils, monocytes, and mast cells. Additionally, lymphocyte subsets (B, CD8, CD4, $\gamma\delta$ T, natural killer, natural killer T, and group 2 innate lymphoid cells) were found. We also found three new subtypes of resident macrophages in the cornea. Trajectory analysis suggested a differentiation pathway from monocytes to conventional dendritic cells, resident macrophages, and LCs. Intercellular communication network analysis using cord diagram identified amyloid precursor protein, chemokine (C–C motif) ligands (2, 3, 4, 6, 7, 9, and 12), *Cxcl2*, *Mif*, *Tnf*, *Tgfb1*, *Igf1*, and *Il10* as prominent ligands involved in these interactions. Sexually dimorphic gene expression patterns were observed, with male myeloid cells expressing genes linked to immune regulation (*Egr1*, *Foxp1*, *Mrc1*, and *Il1rn*) and females showing higher expression of antigen presentation genes (*Clic1*, *Psmb8*, and *Psmb9*). Finally, immunostaining confirmed the spatial distribution of these cell populations within the cornea.

CONCLUSIONS. This study unveils a diverse immune landscape in the mouse cornea, with evidence for cell differentiation and sex-based differences. Immunostaining validates the spatial distribution of these populations, furthering our knowledge of corneal immune function.

Keywords: cornea, heterogeneity, single-cell RNA seq, sexually dimorphic, macrophage

The cornea is the major refractive surface of the eye, playing a crucial role in focusing light onto the retina. It also serves as a primary barrier against external pathogens and injuries.^{1,2} Therefore, understanding the immune landscape of the cornea is vital for understanding immune surveillance and developing therapeutic strategies for corneal diseases and injuries. Several types of immune cells, including neutrophils, dendritic cells, macrophages, natural killer cells, T cells, and innate lymphoid cells, have been detected in the cornea using flow cytometry or immunostaining with antibodies to conventional lineage markers.^{1,3,4} However, detailed characterization of these cells at the single-cell level, including their roles in immune homeostasis and the cytokines and growth factors they produce, remains to be fully elucidated.

Recent advances in single-cell RNA sequencing (scRNA-seq) provide an opportunity to characterize the heterogeneity

of immune cells in an unbiased manner.^{5,6} However, most scRNA-seq studies from the cornea have focused on limbal stem cells, nonmyelinated Schwann cells, limbal region immune cells, or the whole cornea, of which the largest percentage of cells are nonimmune (e.g., epithelial cells, keratocytes, and endothelial cells).^{7–12}

Studies have shown that males and females exhibit distinct immune cell profiles and responses owing to genetic and hormonal influences.^{13–15} For example, males generally have a greater susceptibility to infections, whereas females are more prone to autoimmune diseases, including dry eye disease.^{16,17} It is reasonable to hypothesize that the cornea, being an immune-privileged site, may also exhibit sex-specific immune characteristics. Therefore, including both male and female subjects in our study is crucial to comprehensively understand the immune landscape of the cornea and to develop targeted therapeutic strategies for both sexes.

To enrich immune cells from the cornea, commonly used techniques include microbead-based isolation and flow cytometry-based sorting. However, the impact of these methods on the total number of immune cells enriched or the total number of immune cell clusters identified has not been compared systematically in the cornea. This comparison is essential to ensure accurate and reliable characterization of the corneal immune cell landscape.

Our study identified distinct immune cell populations and observed sexually dimorphic gene expression patterns in corneal myeloid cell clusters. We also found that the flow cytometry method was more efficient in enriching immune cells as compared with the microbead-based isolation method. These findings pave the way for future research into corneal immune responses, disease susceptibility, and the influence of sex on these factors.

MATERIALS AND METHODS

Animal Protocol and Care

All animal procedures adhered to the ARVO Statement for the Use of Animals in Ophthalmic and Vision Research and were approved by the Institutional Animal Care and Use Committee at Baylor College of Medicine (BCM). Male and female C57BL/6J (B6) mice were purchased from Jackson Laboratories (Bar Harbor, ME, USA). At the time of experiments, the mice were aged 16 to 20 weeks. Mice were housed in a nonstressful environment with controlled temperature (22°C–24°C), humidity (50%–75%), and a 12-hour light/dark cycle with free access to food and water. Mice were allowed to acclimate to the housing conditions for at least 2 weeks before the experiment.

Corneal Cell Suspension and Immune Cell Enrichment

A total of 75 adult mice were used in these experiments. Two experiments involved female mice ($n = 25$ per experiment), and one experiment involved male mice ($n = 25$ per experiment). Only female mice were used for the CD45 microbead experiment. The procedure began with an initial incision on the cornea using a No.15 scalpel (Graham Field Feather, 2975#15). The cornea was then held at the edge with the help of a forceps. Subsequently, a scissor was used to cut around the cornea slowly, ensuring the limbal area was excluded. After excision, corneas were pooled and incubated for 20 minutes in 20 mM EDTA-PBS (Sigma Aldrich, St. Louis, MO, USA [E7889]; Corning, Corning, NY, USA [21-040-CV], pH 7.4) to facilitate enzymatic dissociation. After incubation, the tissues were dissociated mechanically into small pieces using sterile scissors. The resulting fragments were then treated with 0.1% collagenase type IV (Gibco, Grad Island, NY, USA [17104019]) for 1 hour at 37°C with constant agitation to achieve a single-cell suspension. Two strategies were used for immune cell enrichment: magnetic bead separation and fluorescence-activated cell sorting. For magnetic separation, CD45 microbeads (Miltenyi Biotec, Bergisch Gladbach, Germany [Cat.no. 130-052-301]) were used to specifically enrich CD45 cells. The single-cell suspension was incubated with the beads according to the manufacturer's instructions and subsequently loaded onto a MACS column and placed in a magnetic field using a MACS Separator (Miltenyi Biotec). This approach isolated the CD45 fraction enriched for immune cells. Alternatively,

fluorescence-activated cell sorting was used for immune cell isolation. Pooled samples were pretreated with anti-CD16/32 (BD Pharmingen, Franklin Lakes, NJ, USA [553141]) to block Fc receptors and minimize nonspecific antibody binding. After incubation, the cells were stained with anti-CD45 antibodies (BioLegend, San Diego, CA, USA [103138]) to identify leukocytes and an infrared fluorescent viability dye (Life Technologies, Waltham, MA, USA [L10119]) to exclude dead cells. Cell sorting was performed on an Aria-II cell sorter (BD Biosciences) at the BCM cytometry and cell sorting core. A sequential gating strategy was used: initial gate on the forward-scatter area and side-scatter area identified total cells, followed by singlets gates (forward-scatter area vs. FSC-W and side-scatter area vs. SSC-W) and, finally, exclusion of dead cells based on the viability dye. A fluorescence minus one control was used to establish a gate for CD45 cells. This approach allowed for the isolation of highly specific immune cell populations based on CD45 expression.

Library Preparation

Single-cell gene expression libraries were prepared using the Chromium Single Cell Gene Expression 3v3.1 kit (10X Genomics) at the Single Cell Genomics Core, BCM. Initially, single cells, reverse transcription reagents, Gel Beads containing barcoded oligonucleotides, and oil were loaded onto a Chromium controller (10X Genomics) to form single-cell Gel Beads-In-Emulsions (GEMs). Within these GEMs, full-length cDNA was synthesized and uniquely barcoded for each single cell. After GEMs were disassembled, cDNA from individual cells was pooled, followed by cleanup using Dynabeads MyOne Silane Beads. Subsequently, the cDNA was amplified via PCR. The amplified product underwent fragmentation to the optimal size, followed by end repair, A-tailing, and adaptor ligation. The final library was then generated through amplification.

Sequencing of 10x Gem 3'v3.1 Single Cell Libraries

The BCM Genomic and RNA Profiling Core conducted initial sample quality checks using the NanoDrop spectrophotometer and Agilent Bioanalyzer 2100. For quantification of the adapter-ligated library and confirmation of successful P5 and P7 adapter incorporations, the Applied Biosystems ViiA7 Real-Time PCR System and a KAPA Illumina/Universal Library Quantification Kit (Illumina, San Diego, CA, USA [p/n KK4824]) were used. Subsequently, the BCM Genomic and RNA Profiling core performed sequencing of the libraries on the NovaSeq 6000 Sequencing System using the S2 v1.0 Flowcell. Cluster Generation by Exclusion Amplification that involved loading 150 pM of the equimolar pooled library onto one lane of the NovaSeq S2 v1.0 flowcell (Illumina p/n 20012860) following the XP Workflow protocol (Illumina kit p/n 20021664). Amplification was achieved by exclusion amplification onto a nanowell-designed, patterned flowcell using the Illumina NovaSeq 6000 sequencing instrument. PhiX Control v3 adapter-ligated library (Illumina p/n FC-110-3001) was spiked-in at 1% by weight to ensure balanced diversity and to monitor clustering and sequencing performance. The libraries were sequenced according to the 10X Genomics protocol, comprising 28 cycles for read 1, 10 cycles each for the i7 and i5 reads, and 90 cycles for read 2. On average, 251 million read pairs per sample

were sequenced. FASTQ file generation was conducted using bcl2fastq, and QC reports were generated using CellRanger v5.0.1 by the BCM Multiomics Core.

Bioinformatic Analysis of scRNA-Seq Data

Raw sequencing reads in FASTQ format were aligned to the mouse reference genome (mm10, 2020-A) using the Cell Ranger Count v7.1.0 pipeline (<https://cloud.10xgenomics.com>). The pipeline used default settings for alignment, barcode assignment, and UMI counting. Following alignment, the resulting gene expression matrix was subjected to standardized quality control procedures specific to scRNA-seq data using the Seurat 5.0.0 package in R.¹⁸ Briefly, the quality control pipeline began with the raw count feature matrices generated by Cell Ranger. Filtering steps were applied to remove low-quality cells based on gene and mitochondrial content. Cells with either less than 200 or more than 4000 expressed features (nFeature_RNA) and a mitochondrial content (percent.mt) above 10% were excluded from further analysis. To mitigate unwanted sources of variation stemming from mitochondrial contamination, the “vars.to.regress = ‘percent.mt’” function was used.

Clustering, Visualization, and Cell Annotation

We analyzed the scRNA-seq data from male and female samples using the Seurat 5.0.0 package. We first combined the datasets using FindIntegrationAnchors and IntegrateData functions (default setting). After identifying variable features with FindVariableFeatures and scaling the data with ScaleData, we performed a principal component analysis for dimensionality reduction. We then used a graph-based clustering approach within the principal component analysis space, followed by nonlinear dimensionality reduction using Uniform Manifold Approximation and Projection (UMAP) for visualization. Using FindNeighbors, we explored different resolutions (0.1–1.0) and selected 0.5 based on our experience and differential expression analysis of the clusters. Finally, we identified cluster-specific marker genes with FindAllMarkers function. Dot plots were created to display both nonimmune cell exclusion markers (*Krt12*, *Tjp1*, *Aldh3a1*, *Epcam*, *Fxyd3*, *Clrn1*, *Htr1d*, and *Grip1*) and cluster-specific identification markers that we have used previously in our studies to assign annotation to each cluster.^{19–21} To confirm cell identity, we used the Cluster Identity Predictor (CIPR) web tool (<https://aekiz.shinyapps.io/CIPR/>) achieving 100% agreement (Supplementary Table S2).

Cell-Cell Interaction and Gene Pathway Analysis

We used the CellChat tool (<https://github.com/jinworks/CellChat>) to infer and analyze intercellular communication networks between cell types, enabling a quantitative understanding of these interactions. Additionally, gene pathway analysis within each cluster was conducted using Qiagen Ingenuity Pathway Analysis (IPA).

Cell Trajectory Analyses

Trajectory analysis was performed with the R package Monocle 3 (<https://cole-trapnell-lab.github.io/monocle3>). The data were preprocessed by converting it to a format

compatible with Monocle3 and performing dimensionality reduction to capture the most relevant information with clustering and dimensionality reduction. A principal graph within each partition was fit using the learn graph function, and the results were visualized using the UMAP method. Subsequently, the cells were arranged according to their progress through the developmental program. This arrangement, known as ‘pseudotime’, represents the cells’ trajectory along the developmental pathway.

Differential Gene Expression Analysis Using DESeq2

We performed differential gene expression analysis using DESeq2. We started by reading a Seurat object from an RDS file and extracted the count data and metadata. We then created a DESeqDataSet object from the count matrix using a design formula based on the subset variable. After filtering out cells with low counts (less than 10), we computed size factors using the scan package to normalize the data. We then ran DESeq with the likelihood ratio test (LRT), comparing the full design formula to the reduced formula. The results of the DESeq2 analysis were extracted and saved to an RDS file for future use. The final data included average DESeq2 values, log2 fold changes, p-values, and p-adjusted values. Genes with an adjusted p-value of less than 0.001 were considered significant.

Immunofluorescent Staining and Confocal Microscopy

Whole eyeballs from both male and female B6 mice aged 16–20 weeks were freshly excised and fixed in iced cold methanol for 20 minutes at -20°C . Following a wash with PBS, the corneas were carefully excised from the whole eye just central to the limbus using scalpels and scissors. The corneas were then washed with a washing buffer (0.1% Tween in PBS) for 5 minutes. The corneas were then permeabilized with a permeabilization buffer (1% Triton X-100 PBS) for 1 hour, and subsequently blocked with a blocking buffer (0.1% Tween-20, 0.3% Triton X-100, PBS) containing 20% serum for 1 hour followed by overnight incubation with primary antibodies at 4°C using 5% serum in blocking buffer (Supplementary Table S1). The next day, the corneas were washed three times for 5 minutes each with washing buffer, with gentle shaking at room temperature. This was followed by a 2-hour incubation at room temperature with secondary antibodies diluted in blocking buffer, with gentle shaking. After a brief wash, the tissues were incubated with a nuclear staining dye (Hoechst 33342) for 10 minutes at a 1:5000 dilution. The samples were then finally washed three times for 5 minutes each with washing buffer before being mounted on slides and flattened with coverslips. Immunostaining was visualized using a Nikon laser scanning confocal microscope (Nikon A1R MP, Nikon, Melville, NY, USA). The captured images were processed using NIS Elements Advanced Research (AR) software version 5.30.05 (Nikon). Each immunostaining experiment was repeated at least three times (total $n = 6$: 3 males, 3 females) to ensure reproducibility of the data.

RESULTS

Optimizing Immune Cell Isolation and Characterizing the Heterogeneity of the Mouse Cornea Immune Landscape With scRNA-Seq

We compared two methods, CD45 microbeads, and flow cytometry sorting, for isolating immune cells from mouse corneas for scRNA-seq analysis. Following the enrichment of immune cells from single-cell suspensions, we used droplet-based scRNA-seq to evaluate the immune cell phenotype in the mouse cornea in an unbiased manner. The resulting scRNA-seq data was subjected to quality control and preprocessing using the Seurat 5.0.0 package in R. Our quality control steps included removing cells with more than 10 percent of mitochondrial RNA (percent.mt), and visualizing feature plots for percent.mt against nCount_RNA and nFeature_RNA against nCount_RNA. Additionally, we used principal component analysis and included PC1 and PC2, elbow plots, and Jackstraw plots to ensure the quality and robustness of our data (Supplementary Fig. S1A–B). We performed UMAP alignment for the flow cytometry sorted data set to identify and mitigate potential batch effects (Supplementary Fig. S1C). Following quality control filtering, we analyzed 3,431 cells isolated with CD45 microbeads and 9,572 cells isolated with flow cytometry (3,352 from male mice and 6,220 from females). We applied UMAP to identify distinct cell populations (Fig. 1A and Supplementary Fig. S2A). We then conducted a dot-plot analysis using nonimmune cell (NIC) markers (*Krt12*, *Tjp1*, *Aldb3a1*, *Epcam*, *Fxyd3*, *Cln1*, *Htr1d*, and *Grip1*), lineage-specific pan markers (e.g., *Ptprc*, *Itgam*, and *Cd3*), and cluster-specific identification markers that we have used previously.^{19–27} The comparison of these two immune cell enrichment methods, as presented in Table, reveals notable differences between flow cytometry sorting and CD45 microbeads enrichment. Flow cytometry sorting demonstrated superior efficiency in isolating immune cells, with 72.7% and 89.6% of cells identified as immune in female and male samples, respectively. In contrast, CD45 microbeads enriched only 42.0% of immune cells. Nonimmune cell contamination was lower in the flow cytometry-sorted samples, with 27.3% and 10.4% of cells being nonimmune in female and male samples, respectively, compared with 58.0% in the CD45 microbead-enriched sample. These results highlight the enhanced specificity and effectiveness of flow cytometry sorting in enriching immune cells from single-cell suspensions of mouse corneas. At the cluster level, only 29% of the clusters (5/17) were identified as immune cells in the microbead isolated sample as compared with 65% (17/26) by flow cytometry sorting methods (Fig. 1B and Supplementary Fig. S2B). After removing nonimmune cells and cell clusters with fewer than 30 cells, we merged clusters with similar gene expression patterns and identified 14 immune cell clusters with well-defined gene expression patterns (56% myeloid and 44% lymphoid) in the flow cytometry-sorted samples (Fig. 1B, Supplementary Fig. S3 and Supplementary Table S3). The myeloid-identified clusters include cornea resident macrophages (rMPs), Langerhans cells (LC), two clusters of conventional dendritic cells 2 (cDC2 and cDC2^{prolif}), neutrophils, monocytes, and mast cells. The identified lymphoid clusters include B cells, T cells (CD8, CD4, and $\gamma\delta$ T), natural killer, natural killer T cells, and group 2 innate lymphoid cells. These findings reveal that flow cytometry is a more effective method for isolating immune cells from mouse corneas for scRNA-seq analysis.

Additionally, our study demonstrates the diverse immune cell landscape within the mouse cornea, with distinct populations identified based on their gene expression profiles. From this point forward, all subsequent analyses were conducted using the immune cells dataset isolated via flow cytometry.

Corneal Macrophage Subtypes and Differentiation Trajectory

Previous studies have classified corneal macrophages into *Ccr2*[−] and *Ccr2*⁺ populations.²⁸ *Ccr2*[−] macrophages exhibit local proliferation and are present in the cornea as early as embryonic day 12.5 (E12.5), whereas *Ccr2*⁺ macrophages are recruited from blood monocytes during steady-state conditions and after injury or inflammation. Building on this foundation, we used Dick et al.'s classification strategy to further characterize corneal rMPs.²⁹ To identify the subtypes, we initially subsetted the rMP data and conducted reanalyses across various resolutions. The optimal resolution of 0.2 was determined, and the resulting distinct clusters were visualized using a UMAP plot (Fig. 2A). We then conducted a dot plot analysis using specific genes as previously reported by Dick et al., including *Timd4*, *Lyve1*, *Folr2*, and *Ccr2* and divided rMP into three major subtypes in the cornea (Fig. 2B). These three subpopulations include TLF⁺+*Ccr2*[−] subtype (5.39%, expressing *Timd4* and/or *Lyve1* and/or *Folr2* and have low or absent expression of *Ccr2*), TLF⁺−*Ccr2*[−] subtype (29.36%, that do not express *Timd4* and/or *Lyve1* and/or *Folr2* and/or *Ccr2*, but express other markers like (*Ccl2*, *Ccl7*, *Hbegf*, *Ler3*, *Egr2*, and *Pf4*) and TLF⁺−*Ccr2*⁺ subtype (63.35%, which do not express *Timd4* and/or *Lyve1* and/or *Folr2* but express *Ccr2*). To understand the differentiation pathways of corneal immune cells, we performed trajectory analysis using Monocle 3. We focused on the myeloid cell lineage and used two starting points (progenitors): circulating monocytes and TLF⁺+*Ccr2*[−] rMPs (Fig. 2C). The lines represent pseudotime trajectory, indicating the developmental progression order of these cells. Numbers along the trajectory show the relative position, with higher numbers indicating a later stage. This analysis suggests a potential differentiation pathway where monocytes transition into cDC2, which further differentiates into TLF⁺−*Ccr2*⁺ rMP and LC. Notably, TLF⁺+*Ccr2*[−] rMPs can also transition into the TLF⁺−*Ccr2*[−] subpopulation before becoming LC.

Intercellular Communication Networks and Canonical Pathways in Distinct Immune Cell Populations

After the identification of distinct immune cell populations, we used the CellChat R package to investigate intercellular communication between these populations. Seurat objects, containing the gene expression data for each cell, were uploaded into CellChat, a program that leverages a curated database of ligand–receptor interactions to infer potential cell–cell communication based on the gene expression data. The resulting analysis identified the total number of ligand–receptor interactions (Fig. 3A) and the strength of these interactions between different cell types (Fig. 3B). We further explored this by identifying dominant cell populations acting as signal senders and receivers, depicted in a 2D scatter plot (Fig. 3C). Interestingly, the analysis revealed

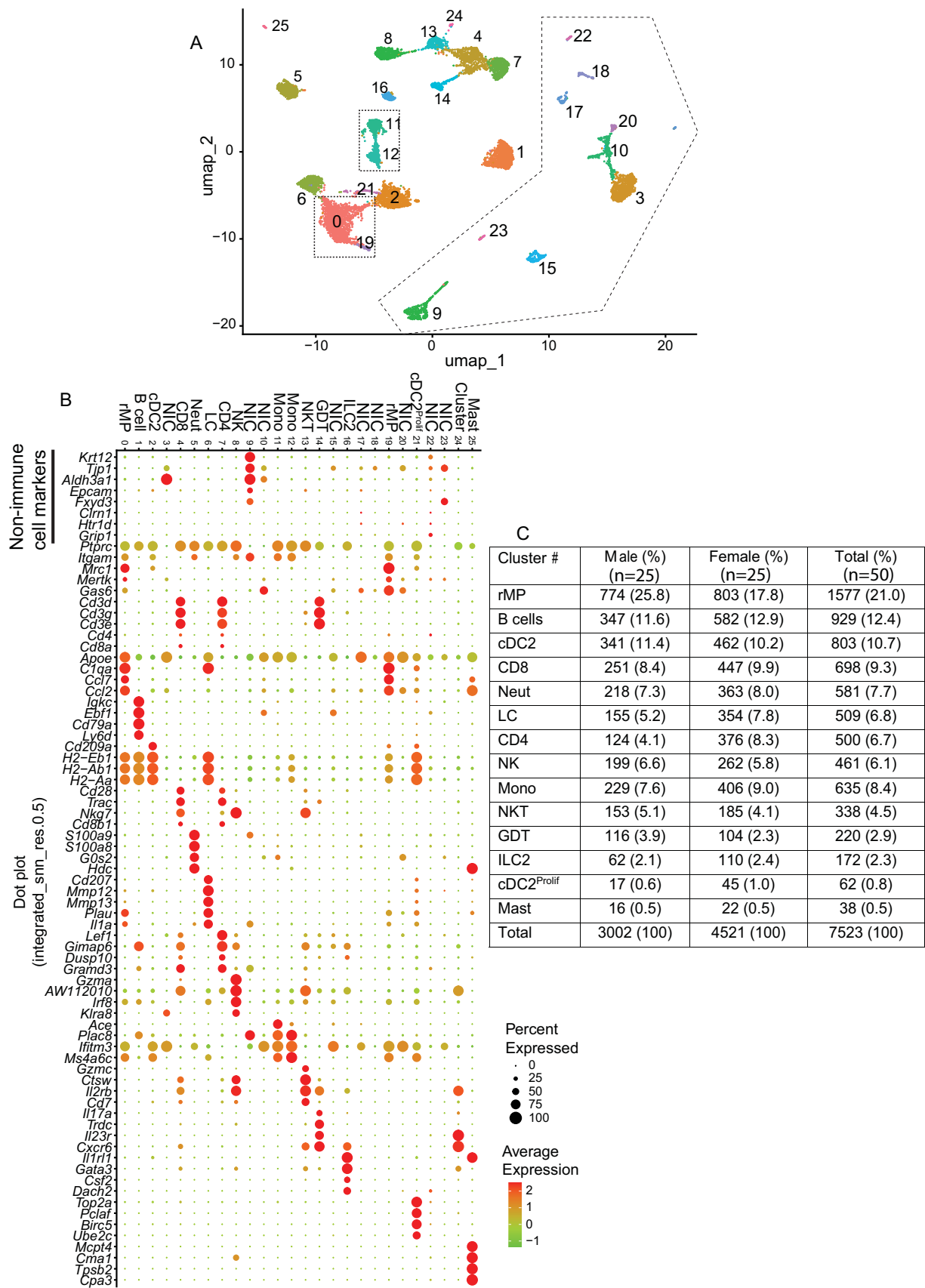


FIGURE 1. Identification of clusters through marker gene expression. **(A)** UMAP plot displays the results of scRNA-seq analysis on CD45 cells isolated from mouse corneas using flow cytometry. Each dot represents a single cell, colored according to its assigned cell type (cluster). The UMAP visualization reveals the distinct clusters of various cell populations present in the mouse cornea. Clusters that express at least one nonimmune cell (NIC) marker (e.g., Krt12, Tip1, Aldh3a1, Epcam, Fxyd3, Cln1, Htr1d, and Grip1) are represented by the dotted polygon. **(B)** The dot plot provides a visual representation of the expression of various markers within each cluster. Identification markers for each

cluster include NIC markers, a pan-immune cell marker (Ptprc), a myeloid cell marker (Itgam), lymphoid cell markers (Cd3d, Cd3g, and Cd3e), and four differentially expressed genes (DEGs). The size of each dot corresponds with the percentage of cells in a specific cluster that express the marker, and the color intensity reflects the average expression level of the marker genes. Clusters exhibiting similar expression patterns (clusters 0, 19, and 11, 12) were combined for further analysis. Clusters expressing nonimmune cell markers, or those with less than 30 cells, were excluded from additional analysis. (C) The table displays the cell count and percentage for each cluster.

TABLE. Comparison of Immune Cell Enrichment Methods: Flow Cytometry Sorting vs. CD45 Microbeads

	Enrichment of Immune Cells Via CD45 Microbeads		Enrichment of Immune Cells Via Flow Cytometry Sorting			
	Experiment #1 (Female)		Experiment #2 (Female)		Experiment #3 (Male)	
	Count	%	Count	%	Count	%
Immune cells	1441	42.00	4521	72.7	3002	89.6
NICs	1990	58.00	1699	27.3	350	10.4
Total	3431	100	6220	100	3352	100

This table compares the two methods for enriching immune cells from mouse cornea samples. The first method uses CD45 microbeads, while the second employs flow cytometry sorting. Results are shown for three experiments, highlighting the count and percentage of immune and NICs obtained by each method.

strong interactions between myeloid (rMP, cDC2, LC, Mono, cDC2^{Prolif}, Mono, and Neut) cells suggesting a potential strong communication network amongst these populations. Further analysis using a cord diagram (Fig. 3D) pinpointed the most prominent ligands involved in these interactions. These included: APP (amyloid precursor protein) chemokine (C-C motif) ligands (2, 3, 4, 6, 7, 9, and 12), C-X-C motif ligand 2, *Mif* (macrophage migration inhibitory factor), *Tnf* (tumor necrosis factor), *Tgfb1* (transforming growth factor beta-1), *Igf1* (insulin-like growth factor 1), and *Il10* (interleukin 10). These ligands interact with their corresponding receptors, such as CD74, C-C motif chemokine receptors (1, 2, and 5), C-X-C chemokine receptor (2 and 4), tumor necrosis factor receptor (–1b and –12a), *Tgfb1R1-R2*, *Igf1r*, and *Il10* receptor A and B. Next, we analyzed the association of each cell cluster with canonical pathways using Qiagen IPA, based on the gene expression data (Fig. 4). The top 50 pathways (selected by *P* value) revealed that rMP cells potentially regulate phagocytosis, anti-inflammatory/regulatory pathways (e.g., CLEAR signaling, IL-10 signaling, macrophage alternative activation, IL-4 and IL-13 signaling, and phagosome maturation) while suppressing proinflammatory pathways (e.g., leukocyte extravasation signaling). Many of these pathways had lower levels of activation in monocytes. Overall, these findings provide valuable insights into the communication landscape between immune cell populations and their potential roles in regulating immune responses.

Sexually Dimorphic Gene Expression Patterns in Corneal Myeloid Cells

Interestingly, male mice exhibited a significantly higher number of rMP cells (*P* < 0.00001), whereas female mice had a greater abundance of CD4 T (*P* < 0.00001) cells (Fig. 1C). Additionally, our CellChat 2D scatter plot identified myeloid cells as the dominant populations acting as both signal senders and receivers (Fig. 3C). Based on these observations, we conducted a differential gene expression analysis of myeloid cells between male and female mice. Our analysis revealed sexually dimorphic gene expression patterns in myeloid cells of the cornea between male and female mice (Fig. 5, Supplementary Table S4). Notably, specific regulatory

genes exhibited distinct expression patterns between the sexes. In male corneas, myeloid cells expressed higher levels of several regulatory genes, including *Egr1*, *Egr2*, *Senp2*, *Maf*, and *Il1rn*. *Egr1* and *Egr2* are transcription factors known to regulate immune cell function and response to stress.³⁰ *Senp2* is an enzyme that modifies protein function, potentially influencing immune signaling pathways.³¹ *Maf* is a transcription factor with diverse roles in the immune system, including lymphocyte development and cytokine production.³² Finally, *Il1rn* encodes the IL-1 receptor antagonist, which helps to regulate inflammatory responses.³³ Conversely, female corneas displayed higher expression of genes involved in protein processing and immune function, including *Psmb8*, *Psmb9*, *Ran*, and *C1qbp*. *Psmb8* and *Psmb9* encode subunits of the proteasome complex, crucial for antigen processing and presentation in the immune response.^{34,35} *Ran*, although primarily responsible for nuclear transport, also plays a role in immune cell activation. *C1qbp* interacts with the complement system, a pathway involved in eliminating pathogens and damaged cells. These findings suggest sexually distinct regulatory mechanisms and protein processing pathways within myeloid cells of the cornea. Further investigation of these differentially expressed genes may provide insights into sex-based differences in corneal immune responses and potential implications for disease susceptibility.

Identification and Localization of Myeloid Cells in the Cornea

Confocal microscopy was used to investigate the distribution of rMPs, cDC2, LC, monocytes, and neutrophils within the cornea. Whole mount cornea staining visualized the spatial distribution of these cell populations, and violin plots present the relative expression levels of their specific markers (Fig. 6). *Mrc1* and *Mertk* expression identified all three subtypes of rMPs throughout the cornea, with a higher concentration in the basal cell layer of the corneal epithelium. Cells expressing CD209b and major histocompatibility complex II (MHC-II) with a characteristic dendritic morphology were identified as cDC2s. Similar to rMPs, cDC2s were found distributed throughout the cornea in the basal cell

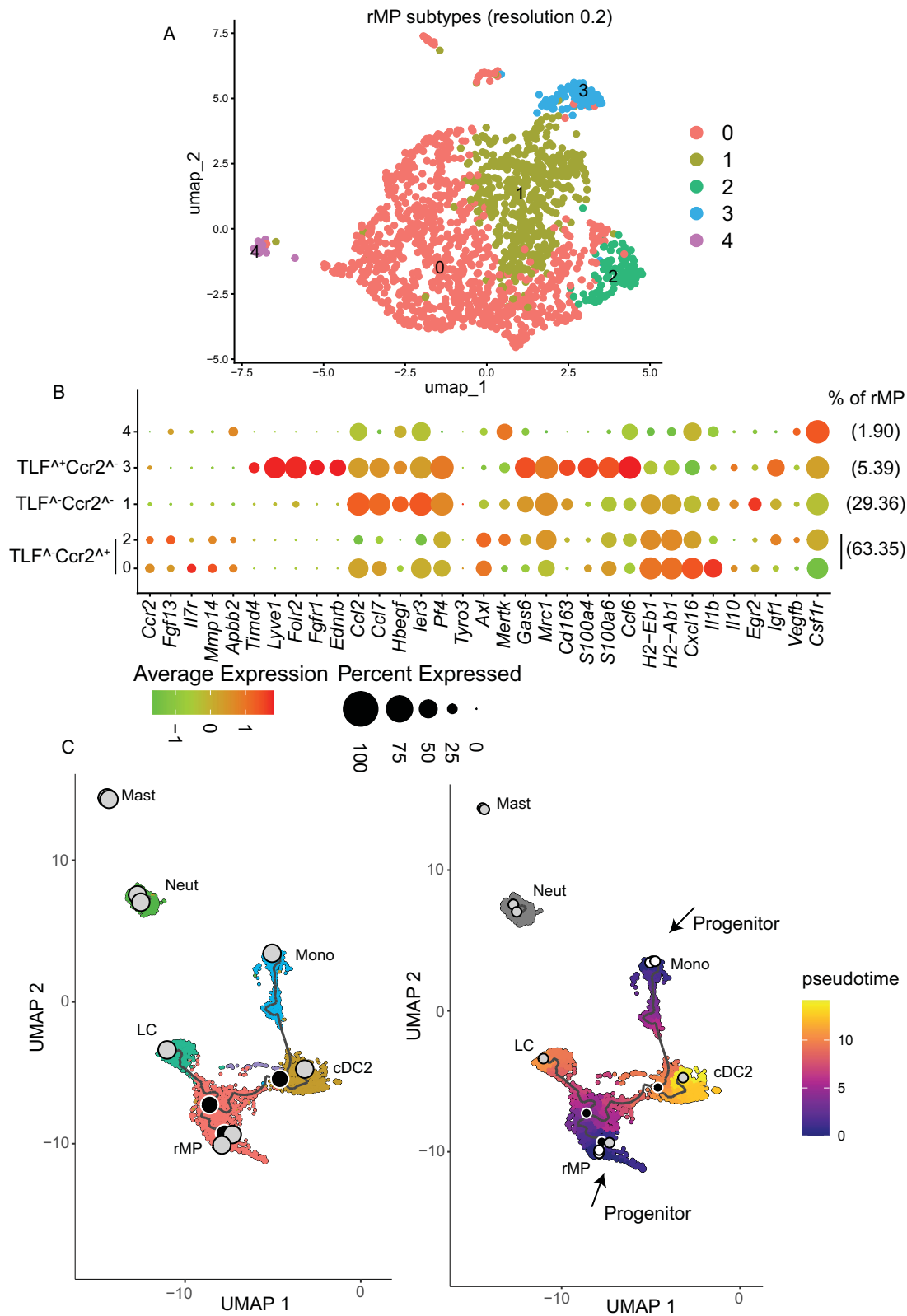


FIGURE 2. Corneal macrophage subtypes and differentiation trajectory. **(A)** UMAP clustering of rMPs into distinct subtypes at an optimal resolution of 0.2. Each color represents a distinct rMP subtype. **(B)** Dot plot displays average expression levels and the percentage of cells expressing key marker genes across different rMP subtypes. The size of each dot indicates the percentage of cells expressing the marker and color intensity reflects the average expression level. Key marker genes include *Timd4*, *Lyve1*, *Fcgr2*, and *Ccr2*. **(C)** Clusters representing different cell types, color coded for identification (*left*), Pseudotime trajectory with a color gradient showing the progression from early (*purple*) to late (*yellow*) stages of cell differentiation (*right*). Progenitors: Starting points of differentiation (circulating monocytes and TLF+CCR2⁻ rMPs). Black circles are branch nodes and grey circles are different outcomes.

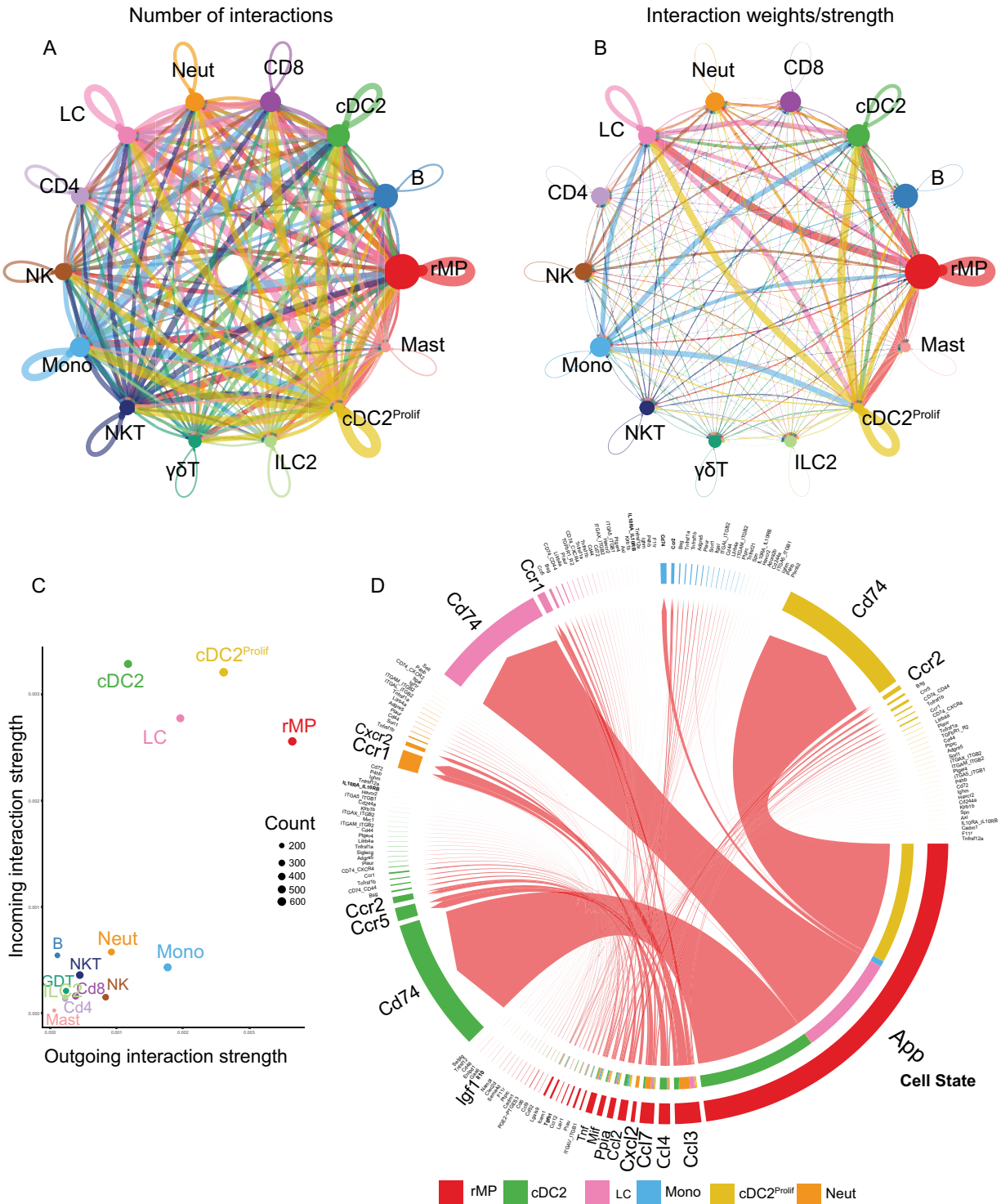


FIGURE 3. Overview of intercellular communication networks in immune cells. **(A)** Total number of ligand–receptor interactions between different cell types. The direction of interactions between different cell types is indicated by color coding. **(B)** Strength of ligand–receptor interactions between different cell types. Thicker lines indicate stronger interaction while thinner lines represent weaker interaction. **(C)** Scatter plot highlighting dominant cell populations acting as signal senders and receivers. **(D)** Cord diagram illustrating the most prominent ligands and receptors involved in intercellular communication between rMP, cDC2, cDC2^{Prolif}, LC, monocytes, and neutrophils.

layer. LC cells, identifiable by their expression of Langerin or CD207 and MHC-II, were primarily located above the sub-basal nerve plexus (SBNP) in the cornea epithelium, with image analysis consistently showing a greater number

of these cells in the peripheral region compared with the center of the cornea. Monocytes, marked by F4/80 and Ly6C staining, were in the deep stroma and the peripheral cornea near the limbal area. Last, neutrophils, detected with

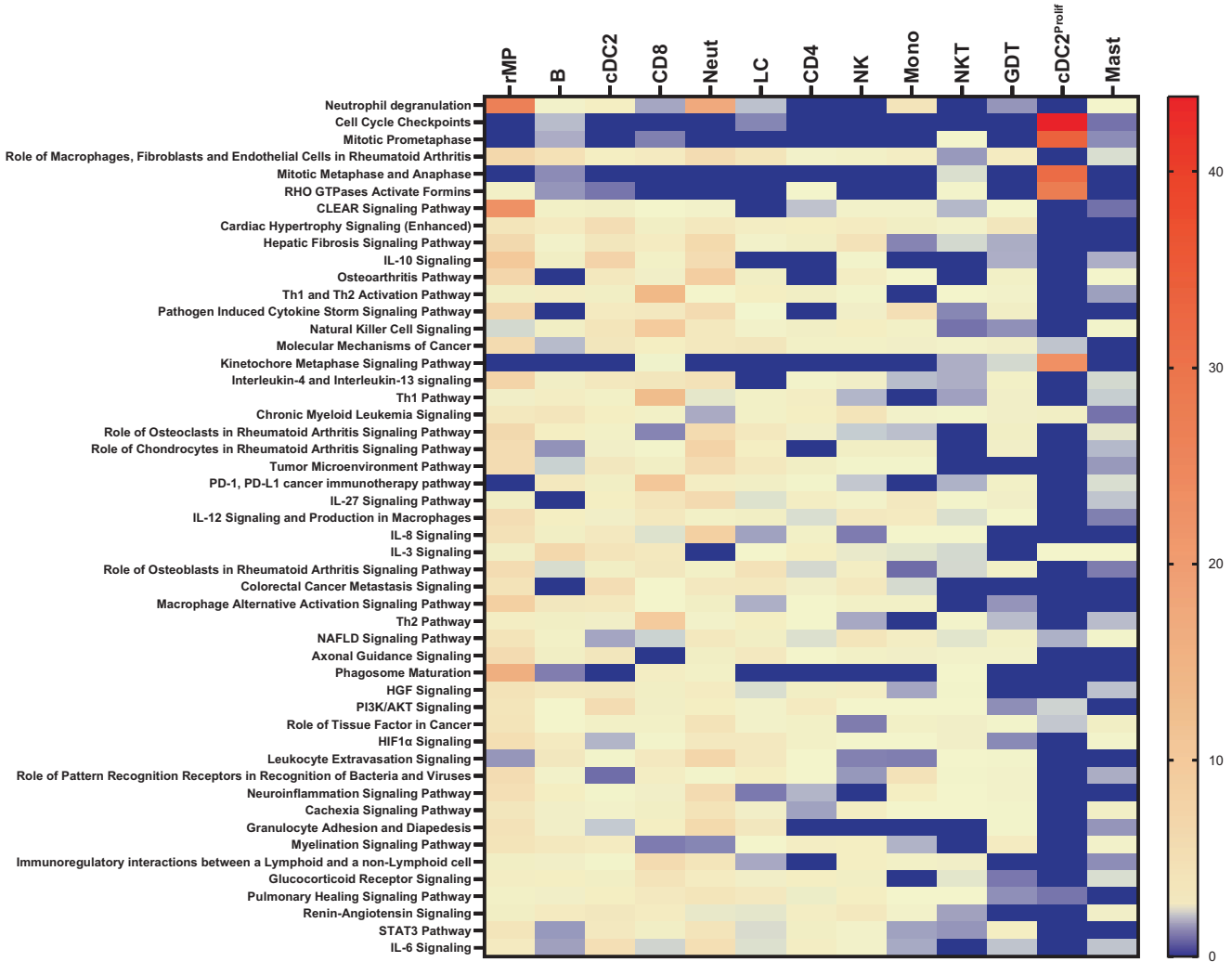


FIGURE 4. Top canonical pathways identified for each cell cluster. Qiagen IPA analysis of gene expression data from each cell cluster identified the top 50 canonical pathways (by P value). The color gradient corresponds with the $-\log(P)$ value.

Ly6G and S100a9 markers, were found in the endothelial layer of the cornea. Additionally, rMPs were discovered to produce important growth factors like IGF1 and VEGFB in the cornea. Through confocal microscopy images, we found that rMP secretes IGF1 in the vicinity of the sensory nerve (Fig. 7A), suggesting a potential role of IGF1 in the local microenvironment around the nerve. VEGFB secretion by rMPs was confirmed at both the scRNA-seq level and the protein level using immunostaining (Fig. 7B). Notably, the IGF1 receptor was found to be expressed on cornea-specific sensory nerves using snRNA-seq data (Fig. 7C, left), as well as on corneal epithelial cells in both mice and humans using immunostaining (Fig. 7C, right).

Reanalysis of scRNA-Seq Data Reveals New Insights Into the rMP Response to Corneal Desiccation Stress

We revisited our previously published cornea immune cells scRNA-seq data, which included control mice and mice exposed to 5 days of desiccation stress (DS5), and applied the new classification strategy of defining rMP into three

subtypes.²⁰ Our data reveal a significant transition from TLF⁺Ccr2⁺ cells, which are predominant under homeostatic conditions, to TLF⁺Ccr2⁻ cells (Fig. 8A). These two subtypes are characterized by the significant upregulation of inflammatory genes such as *Tnf*, *Il1b*, *Cxcl1*, *Nfkb1*, *Nlrp3*, *Cxcl16*, and *Runx1*. Interestingly, the TLF⁺Ccr2⁻ subpopulation maintains a stable proportion throughout the process, suggesting its potential role as a central regulator. This population uniquely expresses both inflammatory cytokines and chemokines (e.g., *Cxcl2*, *Ccl2*, *Ccl3*, and *Ccl12*) to recruit immune cells to the cornea and regulatory genes (e.g., *Apoe*, *Mt1*, *Hmox1*, *Il1rn*, and *Cd163*) that likely dampen inflammation (Fig. 8B, and Supplementary Table S5).

DISCUSSION

Our study aimed to optimize the isolation of immune cells from mouse corneas and characterize their heterogeneity using scRNA-seq. We compared two methods, CD45 microbead and flow cytometry sorting, and found that flow cytometry was more effective in isolating immune cells for scRNA-seq analysis. This result may be because flow cytom-

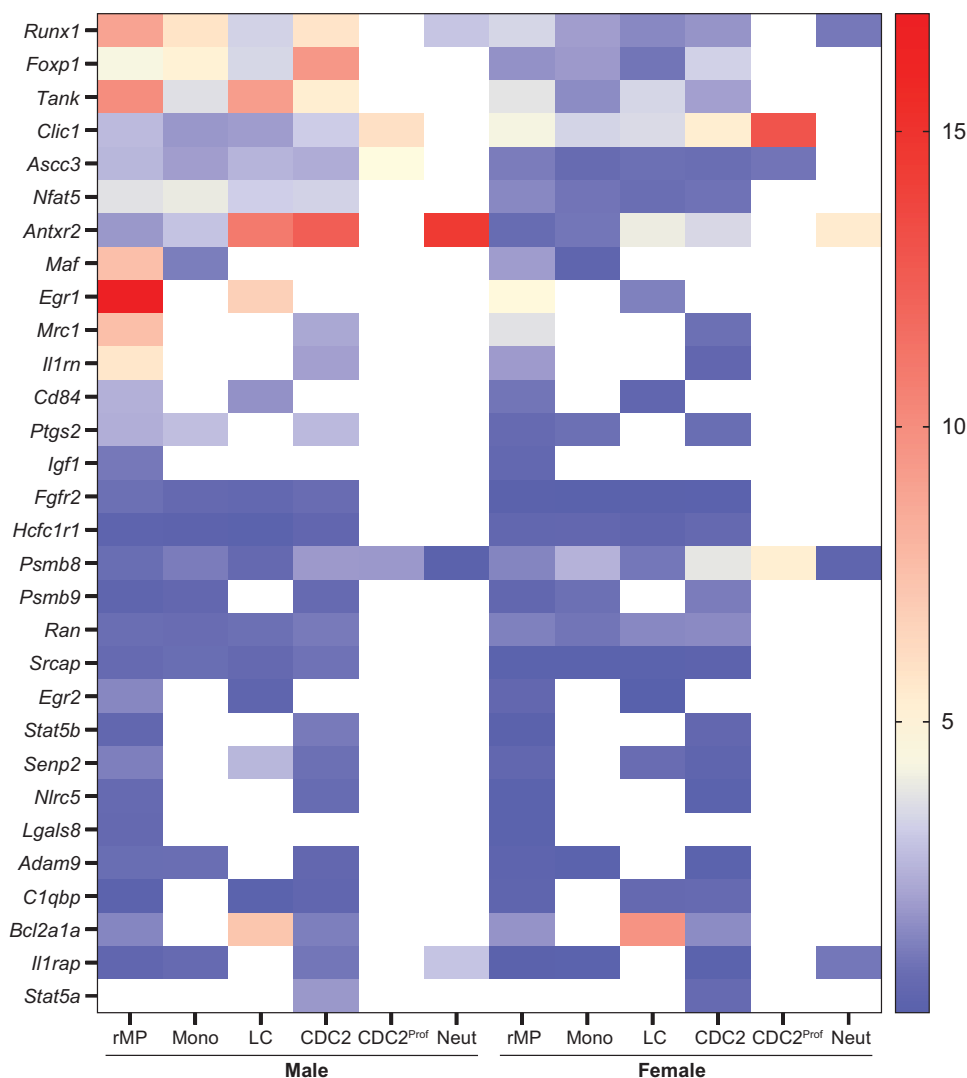


FIGURE 5. Sexually dimorphic gene expression in corneal myeloid cells. This heatmap represents the sex-based differences in gene expression patterns of corneal myeloid cells identified through scRNA-seq analysis. The rows of the heatmap represent different genes, while the columns represent the different myeloid clusters for both males and females (details in Supplementary Table S4). Color intensity denotes the level of gene expression calculated as the average DESeq2 value with warmer colors indicating higher expression and cooler colors indicating lower expression.

etry sorting allows for sequential gating, enabling step-by-step refinement to isolate specific subpopulations, which is not possible in CD45 microbeads. This finding is significant because it provides a more efficient method for future studies investigating the immune cell landscape within the mouse cornea. In line with our recent publication, we discovered that the cornea, under homeostatic conditions, contains 14 immune cell populations. These include rMPs, LCs (we previously reported these as mmp12/13hiMP),²⁰ monocytes, neutrophils, two sets of cDC2, mast cells, and conventional lymphocytes.²⁰ Building on previous studies that classified corneal macrophages into Ccr2⁻ and Ccr2⁺ populations,²⁸ and using a recently published classification strategy,²⁹ we identified three distinct subtypes of rMPs based on the expression of *Ccr2*, *Timd4*, *Lyve1*, and *Folr2* markers. One population (TLF⁺+Ccr2⁻) potentially represents yolk sac-derived macrophages, and the other two populations (TLF⁻-Ccr2⁺ and TLF⁻-Ccr2⁻) might originate from circulating monocytes. This result is consistent with the

findings of Dick et al., who found that TLF⁺+Ccr2⁻ rMPs were maintained through self-renewal with minimal monocyte input, whereas TLF⁻-Ccr2⁺ rMPs (the major group) are entirely replaced by monocytes, while TLF⁻-Ccr2⁻ rMPs receive only a modest monocyte contribution until reaching a defined upper limit.²⁹ The TLF⁺+Ccr2⁻ rMP express genes associated with phagocytosis and apoptotic cell clearance such as *Gas6*, mannose receptor *Mrc1*, and the scavenger receptor *Cd163*. They also express other important genes such as *Igf1* and *Vegfb* (Fig. 2B and Fig. 7). The role of *Igf1* in corneal wound healing and repairing corneal nerves has been well-documented.^{36–40} Similarly, *Vegfb* has been found to play a significant trophic role in the recovery of corneal innervations and trophic functions, particularly in the context of diabetes.^{36,41–43}

Using Cell Chat analysis and IPA, our study also provides valuable insights into the intercellular communication networks and canonical pathways in distinct immune cell populations in the cornea. Analysis of ligand-receptor inter-

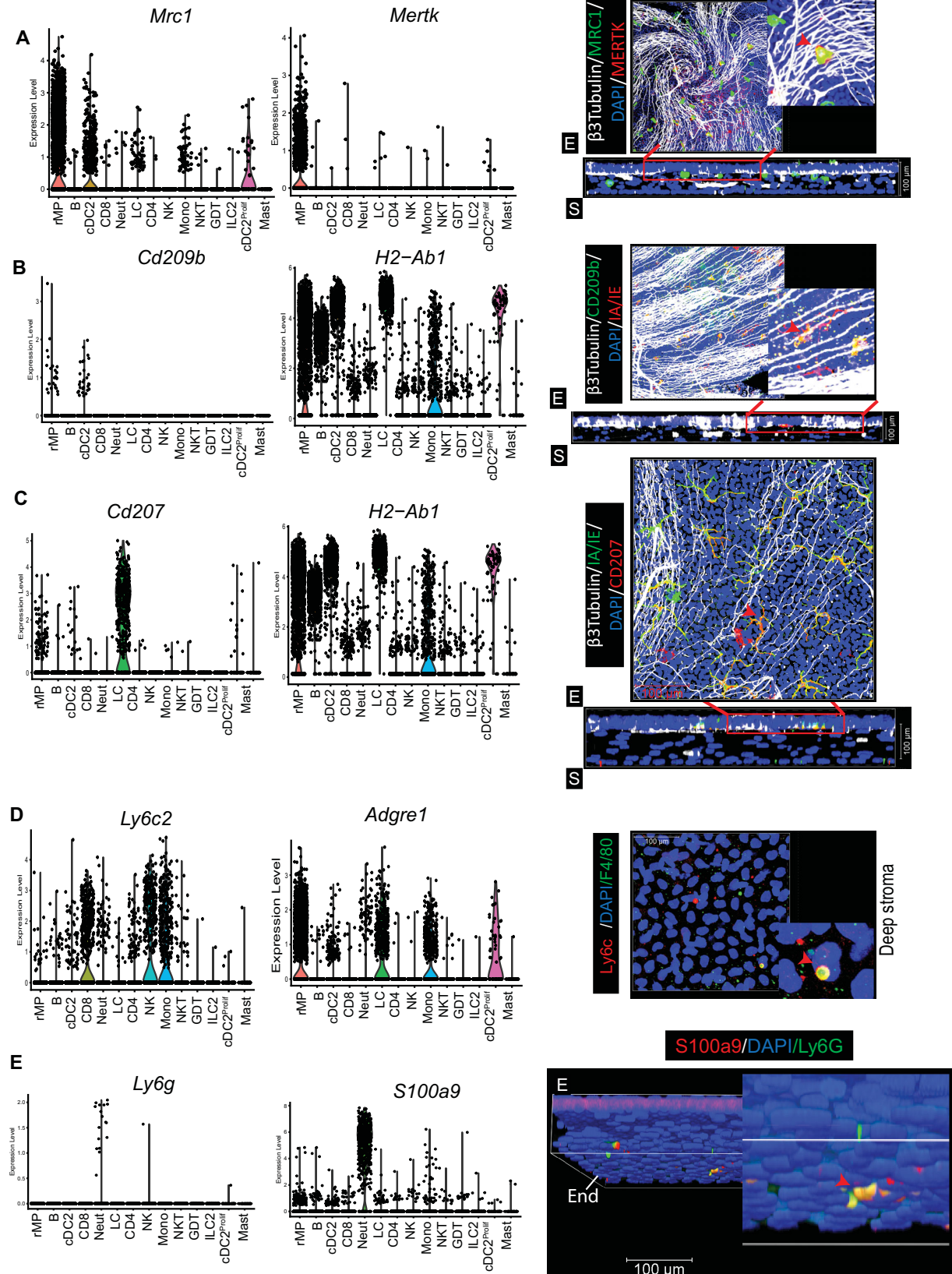


FIGURE 6. Identification and localization of myeloid cells in the cornea. This figure comprises two primary components; violin plots showing the relative expression levels of specific markers for each cell type (*left*), and whole-mount corneal staining that visualizes the spatial distribution of various myeloid cell populations (*right*). (**A**) The markers *Mrc1*(*Cd206*) and *Mertk* are used to identify rMPs. (**B**) cDC2s are characterized by the markers *Cd209b* and *IA/IE* (gene *H2-Ab1*). (**C**) LCs are distinguished by the markers *Cd207* and *IA/IE* (gene *H2-Ab1*). (**D**) Monocytes are identified by the markers *Ly6c* and *F4/80*. (**E**) Neutrophils are marked by *Ly6g* and *S100a9*. Confocal microscopy analysis

elucidates the localization and expression of marker genes in various myeloid cell populations within the cornea. Pseudocolors have been used to enhance the quality of data presentation. The abbreviations used in the figure are as follows: E represents the epithelium, S denotes the stroma, and End signifies the endothelium. Each arrowhead points toward a specific cell, demonstrating the colocalization of the two markers used for identification. The red box delineates the relative area of the cornea for which a 2D image has been presented.

actions revealed a robust communication network among myeloid cells, suggesting coordinated signaling mechanisms that may regulate immune responses in the cornea. Further analysis using a cord diagram pinpointed the most prominent ligands involved in these interactions. These included *App*, *Ccl2*, *Ccl3*, *Ccl4*, *Ccl6*, *Ccl7*, *Ccl9*, *Ccl12*, *Cxcl2*, *Mif*, *Tnf*, *Tgfb1*, and *Il10*. These ligands interact with their corresponding receptors, such as *Cd74*, *Ccr1*, *Ccr2*, *Ccr5*, *Cxcr2*, *Cxcr4*, *Tnfr1a/b*, *Tgfb1r1/r2*, and *Il-10ra/b*, respectively. It is well-established that transforming growth factor β (*Tgfb*) isoforms are profibrotic.^{44,45} However, our scRNA-seq data revealed that, under homeostatic conditions, cornea immune cells are producing *Tgfb1* only not *Tgfb2* (data not shown). This finding suggests that *Tgfb1* might play unique roles in preserving corneal homeostasis, which is distinct from its usual fibrosis-related functions. In addition, our analysis of each cell cluster's association with canonical pathways revealed that rMP might play a suppressive role in inflammation through pathways like IL-10 signaling and macrophage alternative activation. These findings suggest a critical role for these cells in maintaining corneal homeostasis and response to inflammation or injury.

We noticed distinct differences in immune cell populations based on sex. Male mice predominantly have myeloid cells, especially rMP, whereas female mice mainly have lymphoid cells, particularly CD4 T cells. Given that our cell chat analysis pointed to myeloid cells as the most active group, we delved into the differential gene expression patterns within these corneal myeloid cells. Notably, we identified specific regulatory genes that exhibited distinct expression patterns between male and female corneas. Male cornea myeloid cells differentially express genes such as *Egr1*, *Foxp1*, *Mrc1*, *Il1rn*, and *Maf*. These genes act as growth factors and/or are involved in immune regulation and homeostasis.^{46–49} Female cornea myeloid cells express genes (*Clic1*, *Psmb8*, and *Psmb9*) involved in antigen presentation and the activation of key transcription factors, which include interferon regulatory factor 3 and nuclear factor- κ B, which both have roles in autoimmune diseases such as Sjögren's disease.^{50–52} This sexual dimorphism in gene expression may have significant implications for understanding sex differences in corneal immune responses and disease susceptibility. Sexual dimorphism in gene expression patterns has also been reported in previous studies.⁵³ Further investigation of these differentially expressed genes and their functional implications will be crucial for a more comprehensive understanding of corneal immunity. Our study used confocal microscopy to explore the distribution of myeloid cells populations within the cornea. rMPs identified by MRC1 and Mertk were noted throughout the cornea and mostly located in the basal or sub-basal cells layer below the SBNP of the corneal epithelium. Similarly, cells expressing CD209b and MHC-II with a characteristic dendritic morphology were identified as cDC2s and were found distributed throughout the cornea and are also located in the basal cell layer of the epithelium below SBNP. LCs, identifiable by their expression of Langerin or CD207 and MHC-II, are located primarily above the SBNP in the cornea

epithelium, predominantly in the peripheral region. Monocytes, marked by F4/80 and Ly6C staining, were seen in the deep stroma. Last, neutrophils, detected with Ly6G and S100a9 markers, were found in the corneal endothelium. The close proximity of immune cells to the nerves in the cornea contributes to the delicate balance required for nerve survival and reinnervation.⁵⁴

To explore the impact of desiccation stress on rMPs subtypes, we reanalyzed our previously published data.²⁰ Notably, after 5 days of desiccation stress, the TLF⁺–Ccr2⁺ subtype (which is predominant under homeostatic conditions) transitions into the TLF⁺–Ccr2[–] population, accompanied by significant upregulation of inflammatory genes such as *Il1b*, *Nfkb1*, *Nlrp3*, and *Cxcl16*. Interestingly, the TLF⁺+Ccr2[–] subpopulation maintains a stable proportion throughout the process, hinting at its potential role as a central regulator. This population uniquely expresses both inflammatory chemokines (*Cxcl2*, *Ccl2*, *Ccl3*, *Ccl12*) to recruit immune cells to the cornea, and regulatory genes (*Apoe*, *Mt1*, *Hmox1*, *Il1rn*, and *Cd163*) that dampen inflammation (Fig 8B and Supplementary Table S5).

Although our study provides valuable insights into the immune cell landscape within the mouse cornea, there are several limitations that should be considered. First, our study was conducted in mice, and although mice are used commonly as a model for human diseases, there may be differences in the immune cell landscape between mice and humans that were not captured in our study. Future studies could investigate the immune cell landscape in human corneas to validate and extend our findings. Second, our study identified three distinct subtypes of rMPs, but the functional differences between these subpopulations remain unclear. Future studies could investigate the functional roles of these subpopulations in corneal homeostasis and disease processes. Third, although we observed sex differences in the immune cell populations, our study did not investigate the mechanisms underlying these differences directly. Future studies could explore how sex hormones or other sex-specific factors influence the immune response in the cornea. Finally, our study revealed a robust communication network among myeloid cells, suggesting coordinated signaling mechanisms that may regulate immune responses in the cornea. However, the functional implications of these interactions in corneal homeostasis and disease processes were not directly investigated in our study. Future studies could explore how these intercellular communication networks influence the immune response in the cornea. In conclusion, our study provides a more effective method for isolating immune cells from mouse corneas for scRNA-seq analysis. We identified 14 immune cell populations within the mouse cornea and observed sex differences in these populations. We also identified three distinct subpopulations of rMPs and revealed a robust communication network amongst myeloid cells. These findings contribute to our understanding of immune cell ontogeny in the ocular microenvironment and provide a foundation for future studies investigating the role of these immune cells in corneal

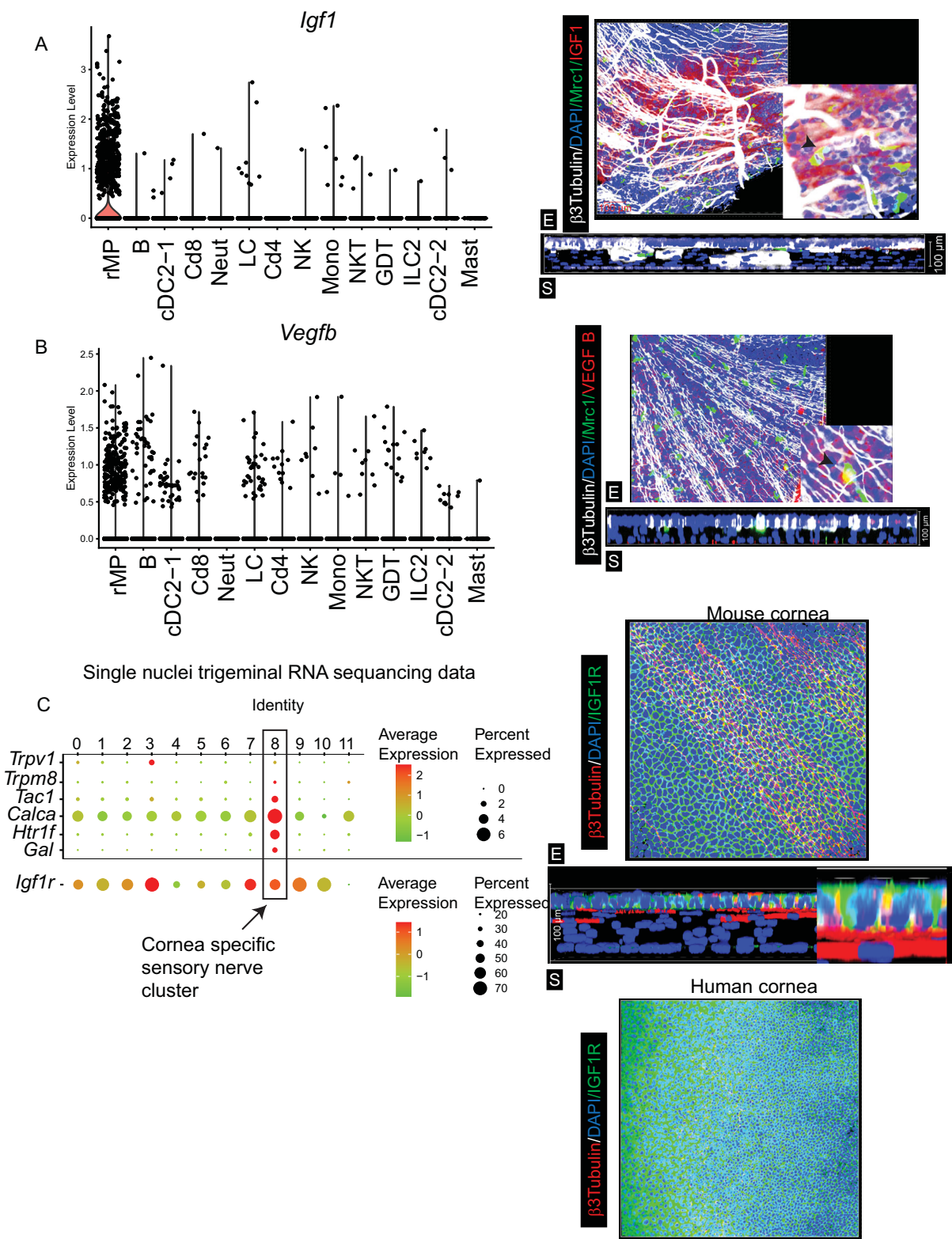


FIGURE 7. Growth factor secretion by rMPs and their interaction with sensory nerves and corneal epithelial cells. **(A)** Violin plot (left) indicating that rMP are expressing Insulin-like growth factor 1 (*Igf1*). The right image confirms this expression at the protein level, as shown by dual labeling and colocalization with *Mrc1* (yellow). Note: *Igf1* also binds with the surrounding matrix. **(B)** Similarly, violin plot (left) shows that rMP also expresses vascular endothelial growth factor B (*VEGFB*), which is confirmed by the right image showing dual staining and colocalization at the protein level. **(C)** Dot plot (left) from trigeminal single nuclei RNA sequencing data (unpublished) shows that cluster expressing the cornea sensory nerve specific genes⁵⁵ are also expressing IGF1 receptor (*Igf1r*), which is confirmed by the right image as colocalization of β 3 tubulin with IGF1R (yellow color). IGF1R is also highly expressed in cornea epithelial cells both in mice and humans. The arrowhead points toward a specific cell showing the colocalization of the two markers used. Pseudocolors were used for data presentation quality. E, epithelium; S, stroma.

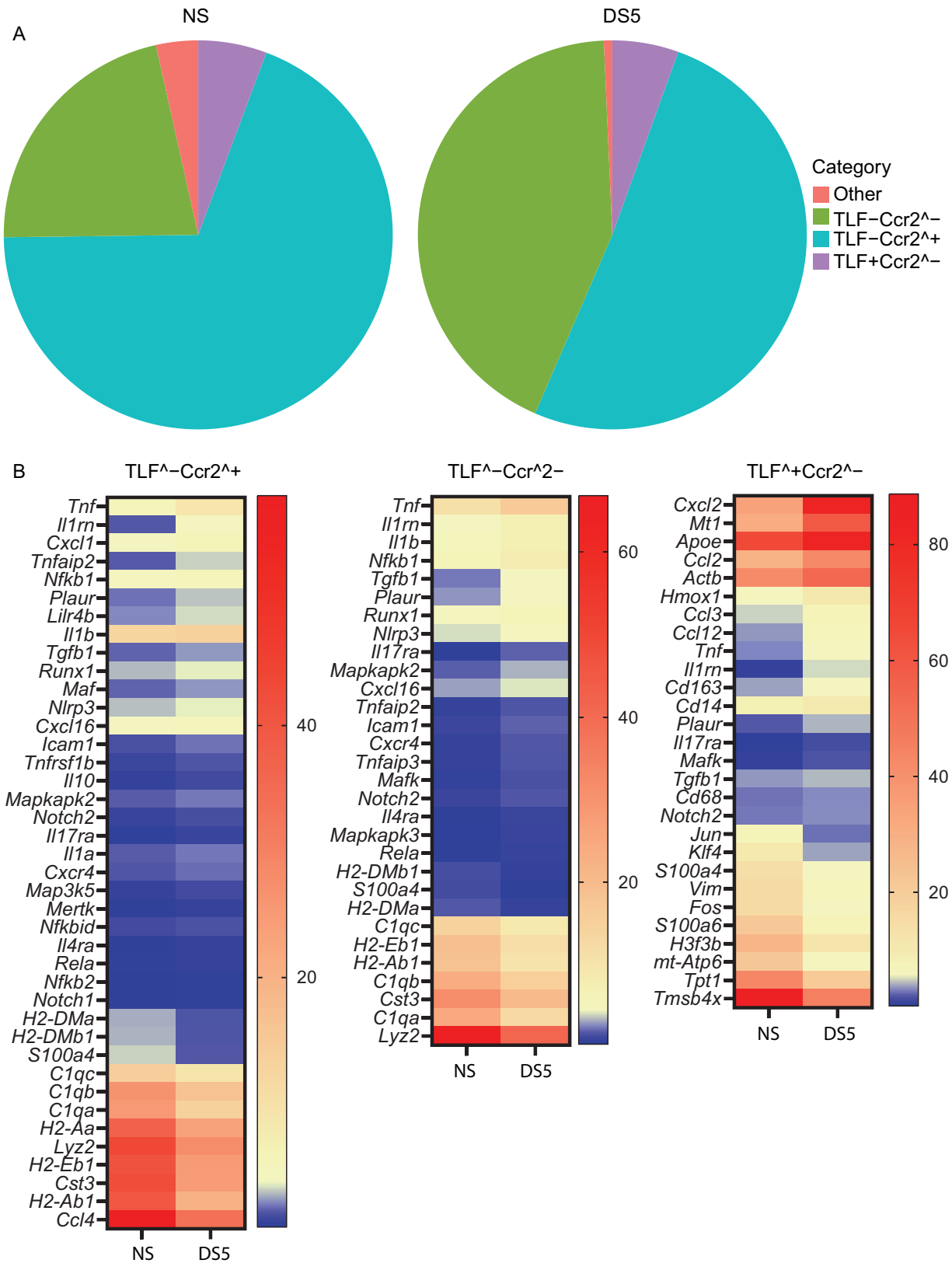


FIGURE 8. rMP subtype distribution and gene expression profile under homeostatic and desiccation stress conditions. **(A)** Pie charts showing the proportion of rMP subtypes (TLF⁺-Ccr2⁺, TLF⁻-Ccr2⁻, and TLF⁺+Ccr2⁻) in control (nonstressful environment) (*left*) and 5 days of desiccation stress (DS5) (*right*) conditions. Each subtype is represented by its corresponding color. **(B)** Heatmaps illustrating the relative expression levels of key genes across the three rMP subtypes. Each heatmap corresponds with a different subtype. The heatmap illustrates the relative differential expression patterns in terms of percentage, with red indicating higher expression and blue indicating lower expression level. All these genes have passed the adjusted *P* value threshold of less than 0.001.

homeostasis wounding and disease processes. Our findings could have implications for the development of therapeutic strategies for corneal diseases and injuries. Further research is needed to validate and extend our findings.

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Data Availability: All raw sequencing data reported have been deposited at this accession URL; https://singlecell.broadinstitute.org/single_cell/study/SCP2610

Code Availability: <https://github.com/jehanalamb82/Characterization-of-the-Mouse-Cornea-Immune-Cell-Landscape-by-Single-Cell-RNA-Sequencing>

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